

: Three-level architecture of OmniCellAgent. **Level 1 (Orchestration):** A Planner decomposes the user query into a **ResearchPlan**; the Executor dispatches each task to the appropriate sub-agent, a Decision node triggers the Re-planner on failure, and the Reporter compiles the final synthesis with citations. **Level 2 (Sub-Agents):** Five specialised sub-agents—OmicMiningAgent, BioMarkerKGAgent, PubMedResearcher, ScientistsAgent, and GoogleSearcher—are invoked in parallel or sequentially by the Executor. **Level 3 (Expert Panel):** Inside ScientistsAgent, an Expert Router (PI) performs structured LLM-based routing to four domain experts (Genomics, Neuroscience, Longevity/Biostatistics, Bioinformatics), each optionally backed by a RAG knowledge base; a PI Synthesis node integrates expert perspectives and returns a unified result.

